

Combinatorics – 7

Extremal set systems and additive combinatorics

Homework

24 сентября 2026 года

Theoretical minimum. Kővári–Sós–Turán: if no s vertices of the part U ($|U| = m$) of a bipartite graph have t common neighbours in W ($|W| = n$), then $e \leq (t-1)^{1/s} mn^{1-1/s} + (s-1)n$, by counting s -stars and Jensen; so a $K_{s,t}$ -free graph has $O(n^{2-1/s})$ edges. The shift graph on pairs $(i, j) \sim (j, k)$ has chromatic number $\lceil \log_2 n \rceil$ (binary digits for the upper bound, the sets $S_i = \{c(i, j) : j > i\}$ for the lower one). Harris–Kleitman: two up-sets (or two down-sets) in $2^{[n]}$ satisfy $|\mathcal{A} \cap \mathcal{B}| 2^n \geq |\mathcal{A}| |\mathcal{B}|$, an up-set and a down-set the reverse. Frankl’s union-closed conjecture is open; Erdős–Ko–Rado is in Combinatorics – 5. Cauchy–Davenport: $|A + B| \geq \min\{p, |A| + |B| - 1\}$ in $\mathbb{Z}/p\mathbb{Z}$; in a finite abelian group $|A| + |B| > |G|$ forces $A + B = G$, and $|A + A| = |A|$ only for cosets of subgroups (affine subspaces in $\mathbb{F}_2^n$). Covering the pairs of a v -set by k -sets needs at least $\lceil \frac{v}{k} \lceil \frac{v-1}{k-1} \rceil \rceil$ blocks; Steiner triple systems need $v \equiv 1, 3 \pmod{6}$. Midpoints and doubling are controlled by the 2-adic valuation: from a and $a + 2^k d$ repeated midpoints give every $a + jd$, $0 \leq j \leq 2^k$.

1. Let $\mathcal{F}$ be a finite union-closed family of sets. Prove that if $\mathcal{F}$ contains a one-element set $\{x\}$, then x belongs to at least half of the members of $\mathcal{F}$, and that if $\mathcal{F}$ contains a two-element set $\{x, y\}$, then x or y belongs to at least half of the members. (Both are special cases of Frankl’s conjecture.)
2. Let p be a prime number and let A be a subgroup of the multiplicative group $\mathbb{F}_p^*$ of the finite field $\mathbb{F}_p$ with p elements. Prove that if the order of A is a multiple of 6, then there exist $x, y, z \in A$ satisfying $x + y = z$. (VJIMC 2014)
3. Let p be a prime and $a_1, \dots, a_{p-1}$ non-zero residues modulo p , not

necessarily distinct. Using the Cauchy–Davenport theorem, prove that every residue modulo p is the sum of some of the a_i (the empty sum being 0), and show that $p - 1$ cannot be replaced by $p - 2$. Prove also that if $A \subseteq \mathbb{Z}/p\mathbb{Z}$ has $k \geq 2$ elements, then $hA = \mathbb{Z}/p\mathbb{Z}$ for every $h \geq \frac{p-1}{k-1}$.

4. Prove the Harris–Kleitman inequality (Theorem 3 of the class sheet) and deduce Kleitman’s theorem: the union of k intersecting families of subsets of $[n]$ has at most $2^n - 2^{n-k}$ members. (Show first that a maximal intersecting family is an up-set with exactly 2^{n-1} members.) Give an example showing that the bound is attained.
5. Prove that n points in $\mathbb{R}^3$ determine at most $2^{-2/3}n^{5/3} + n$ pairs at distance exactly 1: show that the unit-distance graph contains no $K_{3,3}$ and apply the Kővári–Sós–Turán theorem. Then show that in $\mathbb{R}^4$ there are configurations of n points with $\lfloor n^2/4 \rfloor$ unit distances. (The planar bound $O(n^{3/2})$ is in Combinatorics – 6.)
6. Let S be a set with 10 distinct elements. A set T of subsets of S (possibly containing the empty set) is called union-closed if, for all $A, B \in T$, it is true that $A \cup B \in T$. Show that the number of union-closed sets T is less than 2^{1023} . (ICMC 2021)
7. Find all positive integers n for which there exists a family $\mathcal{F}$ of three-element subsets of $S = \{1, 2, \dots, n\}$ satisfying the following two conditions: (i) for any two different elements $a, b \in S$, there exists exactly one $A \in \mathcal{F}$ containing both a, b ; (ii) if a, b, c, x, y, z are elements of S such that $\{a, b, x\}, \{a, c, y\}, \{b, c, z\} \in \mathcal{F}$, then $\{x, y, z\} \in \mathcal{F}$. (IMC 2003)
8. A set C of integers is *sum-free* if there are no $a, b, c \in C$ (not necessarily distinct) with $a + b = c$. Prove Erdős’ theorem: every set B of n non-zero integers contains a sum-free subset with more than $n/3$ elements. (Hint: for a prime $p = 3k + 2$ larger than all $|b|$, $b \in B$, the residues $k + 1, \dots, 2k + 1$ form a sum-free set modulo p ; multiply B by a random non-zero residue.)

9. Call a subset S of $\{1, 2, \dots, n\}$ *mediocre* if it has the following property: whenever a and b are elements of S whose average is an integer, that average is also an element of S . Let $A(n)$ be the number of mediocre subsets of $\{1, 2, \dots, n\}$. (For instance, every subset of $\{1, 2, 3\}$ except $\{1, 3\}$ is mediocre, so $A(3) = 7$.) Find all positive integers n such that $A(n+2) - 2A(n+1) + A(n) = 1$. (Putnam 2009 B3)
10. Let $\mathbb{Z}_{>0}$ be the set of positive integers. Find all nonempty subsets $M \subseteq \mathbb{Z}_{>0}$ satisfying both of the following properties: (a) if $x \in M$, then $2x \in M$; (b) if $x, y \in M$ and $x + y$ is even, then $\frac{x+y}{2} \in M$. (IMC 2025)